



**Figure 6-3:** A four-digit combination bicycle lock has  $10^4$ , or 10,000, possible permutations with repetition (photo courtesy of Shaun Fisher, CC BY 2.0 license).

One way to get permutations with repetition is with *nested loops*—that is, a loop within another loop. The inner loop goes through every element in a set, whereas the outer loop does the same while repeating the inner loop. Creating all possible  $k$ -character permutations, each character selected from a set of  $n$  possibilities, requires  $k$  nested loops.

For example, *nestedLoopPermutations.py* contains code that generates all 3-combinations of {A, B, C, D, E}:

#### **Python**

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```
for a in ['A', 'B', 'C', 'D', 'E']:
    for b in ['A', 'B', 'C', 'D', 'E']:
        for c in ['A', 'B', 'C', 'D', 'E']:
            for d in ['A', 'B', 'C', 'D', 'E']:
                print(a, b, c, d)
```

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